

Website Analytics API - A scalable backend API for website and app analytics built using Node.js, Express, PostgreSQL, Redis, and Docker.

Implements: API Key Management, Event Collection, Analytics Endpoints, Caching, Rate Limiting, Testing, and Swagger Documentation.

Tech Stack: Node.js, Express, PostgreSQL, Redis, Docker, Swagger, Jest.

Setup: Clone repo -> Copy .env.example to .env -> docker-compose up --build -> Access <http://localhost:3000/docs>

Endpoints: /api/auth/register (POST) - register API key, /api/auth/api-key (GET), /api/auth/revoke (POST), /api/analytics/collect (POST), /api/analytics/event-summary (GET), /api/analytics/user-stats (GET).

Environment Variables: DATABASE_URL, REDIS_URL, PORT, API_KEY_TTL_DAYS, REQUESTS_PER_MINUTE.

Deployment: Works with Render or Railway using Dockerfile; add .env variables in platform settings.

Testing: Run npm test (uses Jest and Supertest).

Judging Criteria: Code quality, scalability, completeness, deployment, analytics correctness.

Deliverables: GitHub repo with code, README, Dockerfile, live deployed URL, and Swagger docs.